Game Theory

Strategic Thinking and the Beauty Contest Game

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Before any definitions

Choose a number in

$$[0, 100].$$

The winner is the student whose number is closest to

$$\frac{2}{3} \times \text{average of all chosen numbers.}$$

Question. What number would you choose?

Rules

For each player $i = 1, \ldots, n$:

$$x_i \in [0, 100].$$

After all choices are submitted, compute

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

The target is

$$T(x) = \frac{2}{3}\bar{x}.$$

Player i wins if $|x_i - T(x)|$ is minimal.

How did you chosse?

1. Why did you choose your number?
2. What did you expect others to choose?
3. Did your answer depend on what you think others think?
4. Did anyone choose 0? Why or why not?

Question. Is this an optimization problem or a strategic problem?

One round of reasoning

Suppose a player thinks:

"The average will be about 50."

Then the target will be about

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So a natural choice is

$$33.$$

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Question. But what if everyone reasons this way?

Iterated reasoning

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$$50 \longrightarrow 33 \longrightarrow 22 \longrightarrow 15 \longrightarrow 10 \longrightarrow 7 \longrightarrow \cdots \longrightarrow 0$$

Levels of reasoning

Type	Belief about others	Choice
Level 0	no strategic adjustment	50
Level 1	others are Level 0	33
Level 2	others are Level 1	22
Level 3	others are Level 2	15
$\vdots$	$\vdots$	$\vdots$
Limit	infinite mutual reasoning	0

Question. How many levels of reasoning did you use?

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So we may restrict attention to

$$[0, 66.67].$$

Repeat the argument

Once all players choose in $[0, 66.67]$,

$$\bar{x} \leq 66.67 \quad \Rightarrow \quad T(x) \leq \frac{2}{3} \cdot 66.67 = 44.44.$$

So we reduce the possible set again:

$$[0, 100] \to [0, 66.67] \to [0, 44.44] \to [0, 29.63] \to \cdots$$

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$$[0, 100(2/3)^k].$$

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As $k \to \infty$, only 0 remains.

What exactly have we shown?

The reasoning uses two assumptions:

1. Players are rational.
2. Rationality is commonly understood, at all orders.

That is, I am rational, I know that you are rational, I know that you know that I know that you are rational, and so on.

Question. Is rationality alone enough?

What happens in reality?

Theory predicts:

$$0.$$

But in first-time classroom experiments, the winning number is usually much larger.

This is not a failure of mathematics.

It tells us which assumptions are doing the work.

Why “beauty contest”?

Keynes compared financial-market reasoning to newspaper beauty contests:

> *The task is not to choose the face one personally finds prettiest, but to guess which face the average opinion will judge prettiest.*

In strategic environments, we often predict others’ predictions.

A newspaper beauty contest



Newspaper contests made Keynes' analogy vivid.

The strategic point is not beauty. It is the recursion:

What do others think others will think?

Image source: Michigan State University Libraries, Chicago Tribune archive, 1933.

Generalization

Replace $2/3$ by p.
Players choose $x_i \in [0, 100]$ and the target is

$$T_p(x) = p\bar{x}.$$

Case	Prediction
$0 < p < 1$	unique equilibrium: all choose 0
$p = 1$	every common choice is an equilibrium
$p > 1$	boundary effects; reasoning moves upward

A sequential version

Now suppose three players move in order.

1. Player 1 chooses x.
2. Player 2 observes x and chooses y.
3. Player 3 observes (x, y) and chooses z.

The target is still

$$\frac{2}{3} \cdot \frac{x+y+z}{3}.$$

Question. What will players do in this situation? Do you still expect them to play 0?

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